

Standard Form & Indices

Test 5 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 3^9 (1)

A2. 8^8 (1)

A3. 6^9 (1)

A4. $3^2 \times 3^2 = 3^4$ (1) = **81** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 4^2 = 1 \div 16$ (1) = **1/16** (1)

B3. $1 \div 10^4 = 1 \div 10000$ (1) = **0.0001** (1)

Section C — Writing standard form

C1. 4.7 (1) = **4.7×10^4** (1)

C2. 2.8 (1) = **2.8×10^6** (1)

C3. 7.4 (1) = **7.4×10^{-3}** (1)

C4. 3 (1) = **3×10^{-4}** (1)

Section D — Converting back, and calculating

D1. **720,000** (1)

D2. **0.004** (1)

D3. $5 \times 8 = 40$, $10^4 \times 10^2 = 10^6$ (1) = 40×10^6 (1) adjust to standard form: **4×10^7** (1)

D4. $2 \div 5 = 0.4$, $10^7 \div 10^3 = 10^4$ (1) = 0.4×10^4 (1) adjust to standard form: **4×10^3** (1)

Section E — Powers to know, and estimating roots

E1. **64** (1)

E2. $9^2=81$, $10^2=100$ = between **9 and 10** (1)

Section F — Laws of indices with algebra

F1. **p^{10}** (1)

F2. **q^8** (1)

F3. **r^{15}** (1)

F4. $4 \times 7 = 28$ (1) $h^{2+6} = h^8$ (1) = **$28h^8$**

F5. $5^2 = 25$ (1) $j^{3 \times 2} = j^6$ (1) = **$25j^6$**

Section G — Fractional indices (Higher)

G1. $\sqrt{49} = 7$ (1)

G2. cube root of 1000 = **10** (1)

G3. fourth root of 16 = 2 (1) $2^3 = 8$ (1)

G4. cube root of 8 = 2, $2^2 = 4$ (1) $1 \div 4 = \mathbf{1/4}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.